

Experiences of habit formation: A qualitative study

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Habit formation is an important goal for behaviour change interventions because habitual behaviours are elicited automatically and are therefore likely to be maintained. This study documented experiences of habit development in 10 participants enrolled on a weight loss intervention explicitly based on habit-formation principles. Thematic analysis revealed three themes: Strategies used to support initial engagement in a novel behaviour; development of behavioural automaticity; and selecting effective cues to support repeated behaviour. Results showed that behaviour change was initially experienced as cognitively effortful but as automaticity increased, enactment became easier. Habits were typically formed in work-based contexts. Weekends and vacations temporarily disrupted performance due to absence of associated cues, but habits were reinstated on return to work. Implications for theory and practice are discussed.

Keywords: habit; behaviour change; diet; physical activity; social cognition

Introduction

Health behaviour interventions typically have limited success, with initial gains eroding as people disengage from the recommended behaviours (Jeffery et al., 2000). Habit formation offers a potential mechanism for long-term maintenance (Rothman, Sheeran, & Wood, 2009). Habits are automatic responses to contextual cues, acquired through repetition of behaviour in the presence of these cues (Lally, van Jaarsveld, Potts, & Wardle, 2010). Habitual behaviours are elicited with minimal prior deliberation and persist as long as the cues continue to be encountered, thereby sustaining behaviour change (Lally et al., 2010; Verplanken & Wood, 2006).

In one of the few studies of habit development, volunteers performed a self-chosen health behaviour every day (e.g. “walking for ten minutes after breakfast”). Initial repetitions led to large automaticity gains, which progressively reduced in size with further repetition until automaticity reached a plateau (Lally et al., 2010), but there was variation in the level and timing of the plateau. Further insight into habit development may be gained by using qualitative methods to explore individual experiences of the habit-formation process. The present study documents experiences of habit development in participants on a weight-management programme explicitly based on habit-formation principles.¹

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Method

Weight loss intervention

The intervention centred on a leaflet listing 10 “tips” for dietary and physical activity behaviours conducive to weight loss (see Lally et al. 2008). Tips were designed to promote habit formation through context-dependent repetition (doing the behaviours in consistent contexts). A “tick-sheet” was provided for self-monitoring.

Participants were staff in a large voluntary health organisation. The first 10 respondents to an email advertising the study were invited to a group meeting where the intervention was described and the leaflet was distributed. Inclusion criteria were age ≥ 18 years and body mass index (BMI) ≥ 25 kg/m² (indicating overweight or obesity). Participants attended eight weekly weighing sessions and submitted self-monitoring sheets.

Participants

Participants were three men and seven women, aged 24–59 years (see Table 1). Pre-intervention BMIs ranged from 26.0 to 39.5 kg/m². Self-reported adherence to the tips over the eight week period, based on returned self-monitoring sheets, ranged from 58% to 100%.

Interview procedure and schedule

Semi-structured interviews were conducted at the end of the intervention. This report focuses on habit formation, although the primary purpose of the interviews was to gather feedback to inform refinement of the intervention. Interviews explored participants’ thoughts about the tips, whether and how they were adopted, and problems experienced. Participants were encouraged to speak spontaneously, but an interview schedule was used to ensure these topics were covered.

Table 1. Participant characteristics.

Participant number	Gender	Age (years)	Baseline BMI	Average completion of tips (%) ^a
1	Female	59	29.1	Not reported
2	Female	34	26.6	81.3
3	Male	41	36.2	72.1
4	Female	54	39.5	57.6
5	Male	46	34.9	61.8
6	Female	24	26.0	89.5
7	Female	46	34.0	83.0
8	Female	52	28.2	100.0
9	Female	47	32.9	Not reported
10	Male	37	33.0	61.4
Mean		44	32.0	75.8

Note: BMI, body mass index. ^aAverage percentage of tips adhered to over the eight week period.

Thematic analysis

Thematic analysis of verbatim transcripts was used to identify themes (Joffe & Yardley, 2004). In this article, we describe only themes relevant to the habit-formation process,² that is related to behavioural initiation, repetition, maintenance, automaticity, or cues to habitual behaviour. Relevance was assessed after coding was completed.

Results

Three themes relating to discrete facets of the habit-formation process were identified. Responses from two participants did not refer to these three themes.

Theme 1: Strategies used to aid initiation of behaviour change

Participants used a number of techniques in the early stages to initiate and maintain the behaviours recommended by the tips. For eating behaviour changes, they took preparatory action to ensure that healthy foods were more accessible than unhealthy foods (e.g. *"I made myself start bringing in [to work] fruit and rice cakes so that the chocolate bar and the crisps completely went"*; Female, Participant 7 [F7]). This also aided modification of existing unhealthy habits:

[Previously] I was sitting in front of the telly and having a big bar of chocolate every night [...] [Now when I] sit there watching telly, I'm opening a packet of raisins and eating them, or an apple. (M10)

Preparatory action in the early stages was often cognitively effortful, requiring heightened awareness, self-regulation or conscious planning to avoid temptation to purchase unhealthy foods:

At the start of the programme when you're going [food] shopping, you're trying to pick up things that are better for you than others. It was just [hard] trying to remember stuff that you should be eating rather than the stuff that you want to eat. (F2)

For physical activity, planning often raised awareness of existing patterns, aiding identification of how new behaviours could be inserted into current routines:

I had to say to myself, 'okay, when am I going to do the bulk of the walking?' Initially I had to work out my regular day, how many steps that gave me. And I was quite surprised to see how much walking I did at home. [...] I found that if I walked from [work] to [home] ... [or] if I walked all the way to the shops and back home ... that wasn't actually that difficult to do. [...] That's how I incorporated that into my routine. (F7)

Some actions were not easily integrated into existing routines, and alternative behaviours were selected:

The walking thing, for me ... it just didn't work. It didn't fit in with the job that I'm doing, which is why I started the cycling to the station in the morning and home again. (F2)

Theme 2: Experiences of automaticity development

Participants reported that the cognitive effort required to initiate new healthy behaviours gradually reduced and progressively less forethought was needed.

I didn't have to think about [the recommended behaviours] too much really, they became automatic. [...] They become sort of one routine the more it goes on ... maybe the first week you have to think about things a lot but then the second week it's less. (M3)

Minimisation of deliberation made achieving the tips easier:

Some of [the tips] just worm their way into your brain ... I was aware of the five fruit and vegetables thing beforehand but I often didn't achieve it, whereas now there are very seldom any days where I don't. (F4)

Some behaviours became so familiar ("*pretty much second nature*"; M10) that failure to do them felt anomalous:

Now I actually feel quite strange if I haven't [eaten a salad]. (F7)

Theme 3: Programming effective cues to repeated performance

Participants typically chose to perform the recommended behaviours at certain times (e.g. "*every lunchtime since I started I've been out for a walk*"; F1), or in certain situations, such as arriving at work. The new behaviours were often embedded in existing routines and the predictability, stability and order of office routines was particularly conducive to integration of new behaviours:

I would have my breakfast as soon as I got into the office – that was constant, Monday to Friday. (F2)

By contrast, during leisure time, behaviour patterns were less structured, leading some participants to "come unstuck" at weekends:

Weekend evenings have been an issue over the eight-week program because you go out or you get invited out for a meal with friends and you all have a drink and la la la! So weekends [were] the hardest time to actually stick to them. (F2)

Vacations were also problematic as exposure to situational prompts ceased, although habits were usually reinstated on return to the relevant context:

My last two weeks have been a bit of a disappointment ... I was on holiday and it was just you know, takeaways every night. [...] [But] since I've been back from holiday I've gone straight back onto it. (M10)

Discussion

This study explored experiences of habit development in participants following a habit-based weight loss programme. Three themes relevant to the habit-formation process emerged: Strategies used to aid initiation of behaviour change, experiences of emerging automaticity, and issues surrounding the effectiveness of external cues. While these themes echo insights from previous work (e.g. Neal, Wood, & Quinn, 2006; Tam, Bagozzi, & Spanjol, 2010; Verplanken & Wood, 2006), this is the first study to describe how habit formation is experienced and negotiated in everyday contexts. The findings yield some practical recommendations for effective promotion of healthy habits.

The behaviour change literature notes the need for people to have confidence in their ability to initiate new behaviours (Schwarzer, 2008) and to have positive initial experiences of behaviour change (Rothman, 2000). We showed that, with repetition, the behaviours became more automatic and less effortful. This insight may help to

persuade people to try to adopt novel behaviours because they can be reassured that it will become progressively easier.

We showed that linking behaviours to work-day cues worked well due to their stability and predictability, although this did create some difficulties with maintenance at weekends and vacations. However, most participants took up the habits again in weekdays or on return from vacation. Previous research into habit discontinuation has focused on permanent cessation of cue exposure, such as arises from a residential relocation (Verplanken, Walker, Davis, & Jurasek, 2008). Our findings suggest that even brief discontinuation of cue exposure can (at least temporarily) change behaviour, because habitual responses remain dormant, though habitual performance was resumed when the cue was re-experienced. Habit-formation interventions will be optimised by linking performance to contextual features that are present in a wide range of settings or by encouraging people to form multiple habits that support the same behaviour in different settings (e.g. a weekday habit of cycling to work and a weekend habit of cycling to shops).

Actions may be cued by events (e.g. objects, locations and people), activities (prior actions) or times (Einstein & McDaniel, 2007). Our participants tended to select events (e.g. arrival in the office), reinforcing the utility of event cues in real-world habitual action (McDaniel & Einstein, 2000). However, conceptual distinctions between cues may be blurred in practice, because contingencies develop between cues. For example, Participant 2 was prompted to eat breakfast on arriving at the office, an event cue encountered only at certain times (i.e. on weekday mornings). Habitual action may therefore depend on configurations of multiple cues. Further work to examine potential cue interactions is needed.

There are several limitations to this work. The validity of self-reported adherence is questionable, and we did not quantify habit development. However, most participants spontaneously reported characteristics of habit: Automaticity, the importance of repetition and “feeling strange” for not doing the behaviour (Verplanken & Orbell, 2003). Participants also described the target behaviours becoming *more* habitual gradually, in line with the growth curve characteristic of habit formation (Lally et al., 2010) and the continuum model of habits (Moors & De Houwer, 2006). A second issue was that two participants did not mention habit development at all in their interviews. To avoid prejudging habit development, we did not discuss habits in the interviews unless it was raised by participants, and it is therefore unclear whether these participants had developed habits. Failure of habit development would imply a shortcoming of our intervention, but it would not challenge the validity of insights from the other eight participants. Finally, sample size and selection of participants from a health charity limits generalisability to other populations, especially those less motivated to form healthy habits. Qualitative work in more diverse samples may reveal additional themes. Notwithstanding these limitations, this work represents the first study to investigate individual experiences of habit formation among people attempting to make healthy lifestyle changes, and reveals three facets of habit development that could be useful in the design of behaviour change interventions.

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Notes

1. We do not evaluate intervention efficacy in relation to behaviour or weight change here. An evaluation of the intervention, conducted subsequent to this study in a larger sample, has been reported elsewhere (Lally, Chipperfield, & Wardle, 2008).
2. A more comprehensive description of themes arising from these data is provided by Lally (2008).

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